

Michael Ge

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Work Experience

Auria Space, Greenbelt, MD

Software Engineer Intern

June 2024 – Present

- Optimized existing code to create reusable user interface components using React.JS and JavaScript, adding new features such as a reusable group selection tool and cutting maintenance costs by 30%.
- Built a full-stack feature using JavaScript and C# that allowed users to securely delete files through a React frontend and a RESTful ASP.NET Core backend, making the application more user friendly and efficient.
- Analyzed and debugged issues in a large code base to improve accuracy, resolving more than 20 issues.
- Used C# to improve website security and DB access, resulting in enhanced data protection.
- Participated in code reviews, used Jira and Git for project management, and gained experience with Agile methodologies.

STEAM Your Dreams, College Park, MD

Software Engineer Intern

June 2023 - August 2023

- Developed a player movement engine in Unity/C# for a tile-based game, enhancing user experience.
- Created software to generate randomized obstacles and manage obstacle detection, increasing game challenge.
- Gained experience working collaboratively through Git with other interns.

Projects

Augmented Reality Game for Android: Creature Hunt

- Developed a mobile game in Flutter and Dart for creature hunt using real-world locations. Designed an intuitive interface for tracking progress and displaying the player's location alongside nearby creatures on an interactive map using the flutter_map and geolocator packages.
- Established a connection to a Redis database for secure storage and quick updates of player progress and game data.
- Implement shake detection using sensors_plus package to allow players to catch creatures by shaking their phone, with a notification signaling when a creature is within range.

Parallel Programming in C++/ CUDA: Game of Life

- Developed a GPU-accelerated version of Conway's Game of Life using CUDA to enhance computational efficiency.
- Implemented parallel cell updates using CUDA kernels, leveraging thread blocks to process the grid efficiently.

Java:

- Used object-oriented programming in Java to create a blackjack game, managing game logic such as deck creation and shuffling, card dealing, player actions, and win/loss conditions.
- Used Breadth first, Depth first, and Dijkstra's graph traversal algorithms to create a program that finds the shortest path across a maze.

Education

University of Maryland, College Park, MD

BS in Computer Science

September 2021 – May 2025

Selected Coursework: Web Application Development, Programming Handheld Systems, Parallel Computing, Human-Computer Interaction, Data Science, Artificial Intelligence, Advanced Data Structures, Algorithms.

Technical Skills and Interests

Languages: Java, JavaScript/TypeScript, C++, C#, Python, SQL, Ruby, OCaml, Shell Scripting, Dart

Web and Frameworks: HTML, CSS, React, Flutter, Unity, ASP.NET

Development Tools and Database: Visual Studio Code, Git, Android Studio, Jira, Linux, Postgres, Redis

Software Development Concepts: Data structures, Algorithms, Object-oriented programming, Agile